Activity 01

1. What is the purpose of the code?

a. The purpose of the code is to sort one of four given lists (vectors) in C++.

2. What is the name of the sort algorithm used by Walter?

a. Bubble Sort.

3. Explain how `list 1` will be processed by the program.

a. This list has two pointers initialized adjacent to each other. The timeProcessing is demonstrated below:

i. 5 > 1 → Swapped {1, 5, 4, 2, 8}

ii. 5 > 4 → Swapped {1, 4, 5, 2, 8}

iii. 2 < 8 → Pass

iv. Restart Process until all larger elements 'float' to the end of the list.

4. Explain how `list 2` will be processed by the program.

a. This list also has two pointers initialized adjacent to each other. Processing is demonstrated below:

i. -1 = -1 → Pass

ii. -1 < 0 → Pass

iii. 0 = 0 → Pass

iv. Completes one pass through the array since it is already sorted. Completes in O(n) time.

5. Explain what happens when you input `1.21`?

a. Program immediately exits → No error handling.

6. Explain what happens when you input `two`?

a. Program displays an error message and exits.

7. How many functions are used in the code?
    a. There are two functions; main() and bubbleSort().
8. Run `fixme.cpp` several times and explain some of the weaknesses of this code.
    a. List 1 - 4 are not displayed with how they looked before.
    b. Without viewing source code, the user is unable to tell which sorting algorithm is used.
    c. Unmaintainable in current state.
    d. Limited error handling.
    e. Limited adaptability on larger arrays of data; slow sorting algorithm.
    f. Not required use of vector data type.
9. Also elaborate is how these could be fixed.
    a. Display lists using a printList function.
    b. Limit line length to 85 characters a line.
    c. Display lists in selection list.
    d. Allow the option for the user to enter their own list.
    e. Provide more descriptive error handling.
    f. Choose a different sorting algorithm.
    g. Use list data type.